

DATA VORTEX — ROUND 1 PHASE 2

Logic Explanation

Analytical approach for the selected Easy, Medium and Hard questions.

E3 — Average Engagement by Platform

The query groups posts by platform and calculates the average likes, shares, comments and total engagement for each platform. Posts with missing likes and engagement are excluded from these engagement averages. The results are ordered from the highest average total engagement to the lowest.

SQL concepts used: GROUP BY, AVG(), WHERE, ORDER BY and calculated aggregation.

M1 — Which Locations Generate the Most Engagement?

The users and posts tables are joined using user_id. Posts with missing engagement are excluded. The query then groups records by user location, counts the posts and calculates total engagement using SUM(). Finally, locations are ordered from highest to lowest total engagement.

SQL concepts used: INNER JOIN, GROUP BY, COUNT(), SUM(), WHERE and ORDER BY.

H6 — Find the Most Suspicious High Impact Users

The analysis first calculates the overall average engagement. A second aggregation calculates post count, average engagement and total engagement for every user. The final filtering stage keeps users who have fewer than 10,000 followers, whose average engagement is above the overall average, and who have at least one post where shares are greater than likes.

The EXISTS condition checks whether the user has at least one qualifying post without duplicating users in the final result. The final output is ranked by total engagement.

SQL concepts used: CTEs, aggregation, INNER JOIN, CROSS JOIN, EXISTS, correlated subquery, filtering and ORDER BY.

Data Handling Principle

The SQL analysis respects the Phase 1 cleaning methodology. Missing engagement values are not converted to zero because doing so would fabricate engagement. Only records with known engagement are included in engagement-based calculations.